Common local weights and quartics in 297 and 342 variables

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Abstract

We specify two families of integer quartics by complete quadratic systems and coefficient tables. The first has 297 total variables and 271 quadratic rows; the second has 342 total variables and 314 quadratic rows. Their local conditions of orders 1, 10 and 40 use powers of one rational weight. A seven-equation quadratic presentation of these powers replaces separate rational weight computations. We prove the local extraction statement, including the nonvanishing needed for cleared trace equations, and simultaneous completeness at every finite place and at the real places. Under the stated arithmetic premises, all sufficiently large members of the 297-variable family define the integers in the rationals, and every member of the 342-variable family with parameter $N \geq 27464$ does so. The latter gives an explicitly specified quartic with 4938 nonzero monomials and coefficient height less than $2^{1791283}$.

1 Statements and arithmetic scope

All variable counts include the input coordinate X. Degree is total degree. If P has integer coefficients, $H(P)$ denotes their largest absolute value. For $N \geq 1$ put $A_N = 80989^{2N+1}$. The fixed number A_N is a coefficient parameter, not a polynomial variable.

The coefficient tables `quadratic-family297.json` and `quadratic-family342.json` specify primitive integer quadratic rows $Q_{N,j}$ in respectively 297 and 342 coordinates. Define

$$P_N^{(297)} = \sum_{j=1}^{271} Q_{N,j}^2, \qquad P_N^{(342)} = \sum_{j=1}^{314} Q_{N,j}^2.$$

The complete expanded coefficients are recorded in the files

`polynomial-family297.json`
`polynomial-family342.json`.

Theorem 1.1. *Under the arithmetic premises specified below, the following hold.*

1. *There exists $N_0 \geq 27464$ such that, for every $N \geq N_0$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{296},\ P_N^{(297)}(x, \boldsymbol{y}) = 0.$$

   The proof does not determine a numerical sufficient value of N_0.

2. *For every $N \geq 27464$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \boldsymbol{y} \in \mathbb{Q}^{341},\ P_N^{(342)}(x, \boldsymbol{y}) = 0.$$

   In particular $A_{27464} = 80989^{54929}$ specifies a single such polynomial.

3. *Both families have degree exactly four and content one, with*

$$H(P_N) < 2^{1520}80989^{4N+2} \qquad (N \geq 1).$$

For every $N \geq 1$ the numbers of nonzero coordinate monomials are 4268 and 4938, respectively. The largest coefficient of $P_{27464}^{(342)}$ has exactly 1791283 binary digits and is the coefficient of r^2. More precisely, both families have exact coefficient height $80989^{4N+2}+1$ for $N \geq 23$.

Here are the retained arithmetic premises, stated in full in [3, Section 2]. Let

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d = 3-\rho, \quad E : y^2 = x^3 - d^2x, \quad P = \left(\frac{57-28\rho}{8}, \frac{-660+417\rho}{32}\right).$$

We assume the specified group and height assertions $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for $n \neq 0$; the five-contact height inequality with

$$\mathcal{H} = 2^{110}210^{1800}\left(\prod_{p \leq 76176} p\right)^{19} + 128 < 2^{2093448};$$

the obstruction to negative odd valuations supplied by the normalized pole equations outside $2, 3, 17$, together with their stated realization for a positive input less than 17 with square reduced denominator and the required local squareclasses and valuations at $2, 3, 17$; and the simultaneous prime-pattern and conjugate reduction-order assertions with fixed finite congruences, real approximation, arbitrary fixed avoidance, and the weighted separation property used by the source construction. The selected contact primes are 1 modulo $8 \cdot 3 \cdot 17$. The weighted separation and fixed denominator dilation are justified from the stated analytic estimates in [1]. The present argument uses no stronger prime-selection or pole-realization statement. In particular, every additional integer introduced in the finite preparation is fixed before the contact primes are selected.

For precision, the five-contact radical of $t = u/v$ in lowest terms, $v > 0$, is the product of primes $p > 76176$ having positive odd valuation on one of $v, u, u-v, u-2401v, 32u+343v$. Outside the four exceptional finite contact values, the assumed inequality is $h(t) \leq \mathcal{H}M_5(t)^{19}$. The normalized contact change below adds only $\log 3$ to the relevant height bound. The aggregate trace-height theorem and effective elliptic sum-height theorem used in the two respective families are proved in [4, 5]; their different effectiveness properties account for the distinction in Theorem 1.1. Finite coefficient verification alone proves none of the retained global arithmetic premises.

2 One weight for all required local orders

Write σ for conjugation on $K/\mathbb{Q}$. At an odd rational prime, the valuations of K are unramified and normalized by $v(p) = 1$. Let $a \in \mathbb{Q}^{\times 2}$, let $t = a\lambda$, and put

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad F(t) = \prod_{\theta \in \Theta}(t-\theta), \qquad \delta = t - 1.$$

The retained quadratic system implies

$$F(t)e = (a\delta)^5, \qquad \delta^3 q = e. \tag{1}$$

These are identities among the represented quantities; the actual table expresses their products by quadratic rows. On the domain $\delta \neq 0$,

$$R = \frac{F(t)}{a^5\delta^6}, \qquad q = \frac{1}{R\delta^4}, \qquad q_m = q^{2m-1} \quad (m \geq 1). \tag{2}$$

The contact radical $M(t)$ is the product of primes $p > 76176$ for which $v_p(t-\theta)$ is positive and odd for some $\theta \in \Theta$.

Lemma 2.1. *Every solution of (1) has $F(t) \neq 0$. If $p \mid M(t)$, then $\delta \neq 0$, $v_p(\delta) = 0$, and $v_p(R)$ is odd. In particular $q \neq 0$. If $\delta = 0$, then $M(t) = 1$; if $q = 0$, then $\delta = 0$.*

Proof. Every root of F differs from 1, so $F(t) = 0$ would contradict the first equation, since $a \neq 0$. For $p > 76176$, each nonzero difference between distinct elements of $\Theta \cup \{1\}$ is a p-adic unit. A contact prime thus divides just one factor of $F(t)$ to positive odd order and makes δ a unit. Since a is a rational square, $v_p(F(t)) - 5v_p(a)$ is odd. All prime divisors of the reduced numerators and denominators of $1 - \theta$ are at most 76176, proving $M(1) = 1$. Finally $q = 0$ gives $e = 0$, and the first equation forces $\delta = 0$. □

Lemma 2.2. *Let v be an integer-valued discrete valuation, with $v(0) = +\infty$, and suppose $v(q)$ is odd. For arbitrary positive integers m_j and field elements D_j, the nonzero summands of*

$$T = 1 + \sum_{j=0}^{s-1} q^{(2m_j-1)2^j} D_j^{2^{j+1}}$$

have distinct valuations. Hence $T \neq 0$ and

$$v(T) = \min\bigl(0, 2^j((2m_j - 1)v(q) + 2v(D_j)) : 0 \leq j < s\bigr) \leq 0.$$

If $v(T) = 0$, then T is congruent to 1 in the residue field.

Proof. Each finite nonconstant valuation is 2^j times an odd integer. Its exact power of two divisor is therefore 2^j, so it differs from zero and from every valuation with a different index. The smallest valuation is attained once. The ultrametric inequality is then an equality. If that smallest value is zero, all the nonconstant terms have positive valuation. A zero D_j simply removes its term. □

Proposition 2.3. *Suppose $p > 2$, $\nu = v_p(R)$ is odd, and $v_p(\delta) = 0$. Let $W = w_0 + w_1\rho$ be a product of factors from Lemma 2.2, using q from (2). Suppose the rational quantities w_0, w_1, T_0, e, b satisfy*

$$w_0^2 - 2w_0T_0 - 2w_1^2 = 0, \qquad e\delta b = T_0^2, \qquad e\delta = R^{-1}. \tag{3}$$

Suppose that $b \neq 0$ cannot have negative odd p-adic valuation. Then $\nu > 0$, and at one place $v \mid p$ every test in every factor satisfies $v(D_j) \geq m_j$.

Proof. At every place above p, $v(q) = -\nu$ is odd. Every factor is nonzero and has nonpositive valuation by Lemma 2.2. In particular $W \neq 0$. If $w_0 = 0$, the first cleared equation forces $w_1 = 0$, a contradiction. It follows from that equation that

$$T_0 = \frac{\mathrm{N}(W)}{2w_0} = \frac{1}{\mathrm{Tr}(W^{-1})} \neq 0, \qquad b = \frac{R}{\mathrm{Tr}(W^{-1})^2} \neq 0.$$

Thus no inverse of w_0 or unproved nonvanishing assumption has been used. Put $\ell = \max_{v|p} v(W) \leq 0$. The trace inequality gives

$$v_p(\mathrm{Tr}(W^{-1})) \geq -\ell, \qquad v_p(b) \leq \nu + 2\ell.$$

The valuation of b is odd. The obstruction therefore implies $0 < v_p(b) \leq \nu + 2\ell$, so $\nu > 0$ and $\ell > -\nu/2$. Choose a place attaining ℓ. Each factor has valuation at least ℓ, because all factor valuations are nonpositive and sum to ℓ. Consequently every nonzero test term satisfies

$$2^j\bigl((1 - 2m_j)\nu + 2v(D_j)\bigr) > -\nu/2.$$

Thus

$$v(D_j) > \left(m_j - \frac{1}{2} - \frac{1}{2^{j+2}}\right)\nu \geq m_j - \frac{3}{4}.$$

The last inequality uses $m_j \geq 1$ and the positive odd integer $\nu \geq 1$. Integrality gives $v(D_j) \geq m_j$. The assertion is automatic if $D_j = 0$. □

3 Simultaneous completeness

We now verify all places, including primes dividing new denominators. This step is needed because equality of contact valuations alone would not prove completeness.

Fix an integer input, the scalar parameter, the four integer root labels, and the positive rational square scales $a_1, \ldots, a_r$. The retained point preparation gives nonempty real and finite-place neighborhoods on which all the rational test functions are defined. Choose smaller neighborhoods with compact closure; at each of finitely many prepared places their valuations have a fixed lower bound. Fix these neighborhoods and the scalar preparation before choosing contact primes.

Lemma 3.1. *Let S be a fixed finite set of rational primes containing the prepared places and all required fixed denominators. There is a positive integer D, fixed before prime selection, such that the following holds uniformly for integers u, w with $w \neq 0$ and u a prime outside S:*

$$\lambda = \frac{u}{D^2 w^2}, \qquad t_i = a_i \lambda.$$

For every $p \in S$, all the nonconstant terms of every modified factor can be made arbitrarily small p-adically by the choice of D. We may simultaneously require $D^2\theta/a_i \in \mathbb{Z}$ for every i, θ.

Proof. First clear the finitely many indicated denominators. Increasing $v_p(D)$ then gives

$$v_p(\lambda) = -2v_p(D) - 2v_p(w) \leq -2v_p(D).$$

Choose it so that, for each i,

$$v_p(t_i) < \min\big(0,\ v_p(\theta) : \theta \in \Theta \setminus \{0\}\big).$$

The ultrametric inequality gives

$$v_p(\delta_i) = v_p(t_i), \qquad v_p(F(t_i)) = 5v_p(t_i), \qquad v_p(e_i) = 5v_p(a_i).$$

Therefore

$$v_p(q_i) = 5v_p(a_i) - 3v_p(t_i) = 2v_p(a_i) - 3v_p(\lambda) \geq 2v_p(a_i) + 6v_p(D). \tag{4}$$

This lower bound tends to infinity with $v_p(D)$, uniformly in u, w. For a test assigned order m in position j, the valuation of its term is

$$2^j\big((2m-1)v_p(q_i) + 2v(D_j)\big).$$

The tests have fixed valuation lower bounds in the chosen neighborhoods, so every such expression tends to infinity uniformly. The same argument at a place of K above p uses its normalized extension valuation; ramification at two changes no inequality. There are finitely many scales, places and tests, so one fixed integer D works for all of them. □

The additional divisors of D enter only the fixed avoidance modulus. The coefficient list in the prime pattern is fixed after this choice:

$$u - \frac{\theta}{a_i} D^2 w^2.$$

Repeated coefficients are identified. The coefficient 0, contributed by $\theta = 0$, makes u itself one of the selected primes. The simultaneous selection premise supplies these primes with avoidance of the fixed data, the stated conjugate orders, and the fixed congruences. Thus the hypothesis that u is prime outside S is part of the same prime pattern, not a further independent restriction. A fixed enlargement of D changes neither the exponent in the weighted count nor the permitted slowly decreasing separation parameter.

Proposition 3.2. *The retained scalar and prime preparation supplies simultaneous witnesses for all the modified detectors. No additional lower bound on the scalar parameter N is needed for completeness.*

Proof. Apply Lemma 3.1 after the fixed point neighborhoods and scales have been chosen. At each detector, writing $v_* = D^2 w^2$, the exact factorization is

$$R_i = \frac{\prod_{\theta \in \Theta} (u - (\theta/a_i) v_*)}{v_*^5 \delta_i^6}. \tag{5}$$

Its denominator is a rational square. The selected primes avoid its support and the fixed contact differences. Hence $v_p(R_i) = 1$ at its five selected contacts, and $v_p(R_i)$ is even elsewhere.

At a selected contact prime the designated primary place satisfies $v(D_j) \geq m_j$ by the unchanged point and logarithm preparation. Since $v(q_i) = -1$, every new term has positive valuation there. Thus each factor is 1 modulo the primary prime. At the opposite split place Lemma 2.2 says that every factor either has negative valuation or is 1 modulo the prime. Consequently the contributions to $\mathrm{Tr}(W_i^{-1})$ are 1 and either 0 or 1. The trace is a unit, congruent to 1 or 2 modulo $p > 76176$. The pole input

$$b_i = R_i \, \mathrm{Tr}(W_i^{-1})^{-2}$$

has valuation one at each selected contact.

Outside those contacts, (5) gives even valuation of R_i; hence the valuation of b_i is even, whether positive or negative. In particular its reduced denominator is a square. This argument covers all remaining primes, including divisors of the numerator or denominator of δ_i; no unit assertion about δ_i at such primes is needed.

At the finite prepared places, Lemma 3.1 makes every factor arbitrarily close to 1, and hence makes $\mathrm{Tr}(W_i^{-1})$ arbitrarily close to 2. Moreover the same calculation gives

$$v_p(R_i) = -v_p(t_i) - 5v_p(a_i) = -v_p(\lambda) - 6v_p(a_i),$$

which can be made uniformly large by D. Thus $v_2(b_i) \geq 3$, $v_3(b_i) \geq 1$ and $v_{17}(b_i) \geq 2$ follow after a fixed choice of D. At two choose the closeness of the trace to 2 so that its valuation is exactly one; at three and seventeen it is a unit. The local squareclasses of b_i equal those of R_i. Formula (5), with the selected primes congruent to 1 modulo $8 \cdot 3 \cdot 17$, shows that these are squares in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$.

At both real embeddings, uniformly on the fixed compact point neighborhoods, as $\lambda \to +\infty$,

$$R_i = \frac{1 + O(\lambda^{-1})}{a_i^6 \lambda} > 0, \qquad q_i = a_i^2 \lambda^{-3}(1 + O(\lambda^{-1})).$$

Thus each factor is positive for large λ and tends to 1. A term of order m in position j decays as $O(\lambda^{-3(2m-1)2^j})$. Hence $\mathrm{Tr}(W_i^{-1}) \to 2$ and $b_i \to 0^+$, so $0 < b_i < 17$. The selection premise supplies arbitrarily large λ after D and all fixed congruences have been chosen. The pole-realization premise therefore supplies fresh simultaneous witnesses. Equal scales may reuse identical choices. All rational circuit quantities are defined on these witnesses, which proves the proposition. □

4 The quadratic presentation and exact counts

The powers needed by the local factors have the presentation

$$\begin{gathered} q_2 = q^2, \quad q_4 = q_2^2, \quad cq_4 = q, \quad q_8 = q_4^2, \\ cq_{19} = q_8^2, \quad q_{38} = q_{19}^2, \quad cq_{79} = q_{38}^2. \end{gathered} \tag{6}$$

Every row has degree at most two. On $q \neq 0$, the rows successively force

$$q_2 = q^2, \quad q_4 = q^4, \quad c = q^{-3}, \quad q_8 = q^8, \quad q_{19} = q^{19}, \quad q_{38} = q^{38}, \quad q_{79} = q^{79}.$$

There are no choices of roots or signs in this conclusion. At a consumed contact prime $q \neq 0$ by Lemma 2.1. When $q = 0$, that lemma shows that the contact radical is empty; no division by q is used in soundness in this case. Completeness uses $\delta \neq 0$ and fills all seven rows by their displayed rational values.

For the point and logarithm notation of the source, put

$$A_i = f(U_i)\ (1 \leq i \leq 4), \quad B_i = \sigma f(V_i)\ (0 \leq i \leq 4), \quad C_i = L_i - L_i', \quad I = L_1 - \beta, \quad D_H = f(H),$$

$$G = L_3^2 - L_2(L_2 - 2)L_3L_4 + (L_2 - 2)^2L_4^2 - 1, \qquad \beta = 51X^2 + 2504.$$

Here $f = 1/x$ extends by zero at the identity, $U_0 = H$, $V_0 = H - P$, and the retained exact addition rows give $Q_i = U_i - V_i$. The alternate ratios L_i' already include conjugation. The nine tests A_i, B_i have order one; C_1, I have order ten; and D_H, C_2, C_3, C_4, G have order forty. For an ordered list, its factor is

$$\mathcal{T}(D_0, \ldots, D_{s-1}) = 1 + \sum_{j=0}^{s-1} q^{(2m(D_j)-1)2^j} D_j^{2^{j+1}}.$$

The two-detector system uses

$$\begin{gathered} (D_H, C_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3), \\ (C_4, A_3, B_4), \quad (G, A_4), \quad (A_1, B_1), \end{gathered}$$

and the three-detector system uses

$$\begin{gathered} (D_H, C_1, B_1, B_0), \quad (C_2, I, B_2), \quad (C_3, A_2, B_3), \\ (C_4, A_3, B_4), \quad (G, A_4, A_1). \end{gathered}$$

Each of the sixteen tests occurs exactly once. Their field powers and all the factor and product rows are exactly those specified and verified in [2]. They remain quadratic; only the scalar weights change.

In the two-detector system the shared field powers use ten rational coordinates, while the three-detector system uses eighteen. Each field square is represented by

$$(u + v\rho)^2 = (u^2 + 2v^2) + (2uv)\rho,$$

so it takes two quadratic output rows. The needed fourth and eighth powers of the B_i are linear combinations of existing formal-logarithm coordinates, as proved coefficient by coefficient in [2].

The exact table transformation removes the source coordinates $353, 354, 356$ and their translates by 48 for each detector. The coordinate with source identifier 355 now denotes c. The source coordinates $359, 369, 360$ denote q_{19}, q_{38}, q_{79} respectively. The existing coordinates $352, 361, 362$ denote q, q_2, q_4. A coordinate q_8 is already present in each of the three detectors. In the two-detector system it is added with source identifier $850 + i$, $i = 0, 1$. All other rows and coordinates are retained literally. Consequently the systems have the following sizes:

Height scales	Total coordinates	Quadratic rows	Nonzero monomials
$1, S_0^2$	297	271	4268
$1, S_0^2, (S_0+S_1)^2$	342	314	4938

Here $S = \sum_{i=1}^{4} x(Q_i) = S_0 + S_1\rho$. The monomial counts hold for every $N \geq 1$; they are statements about coefficients after substituting A_N, not additional variables.

4.1 Primitive normalization of the scalar contact product

Let z_j denote a coordinate by its source identifier. The actual contact product is the affine expression

$$\begin{aligned} r = & -103171881284 - 2103220824z_{248} - 21050600z_{487} \\ & - 7629700z_{488} - 45771136z_{489}. \end{aligned}$$

We use r itself as the coordinate in place of z_{489}. This is an invertible rational affine change: its coefficient of z_{489} is nonzero, and every other coordinate, including the input, is fixed. The source coordinate occurs only in its own defining row and the scalar normalizer row. After substitution and primitive normalization, the latter is exactly

$$R_s + 80989^2 r R_s - A_N r = 0.$$

Every equation remains quadratic. Reverse substitution reproduces both source rows exactly up to a nonzero rational scalar, so their rational solution sets correspond bijectively. Taking sums of squares preserves the required conjunction over $\mathbb{Q}$.

In the resulting quartic the coefficient of r^2 is $A_N^2 + 1$. Every other coefficient is either a nonzero constant or a nonzero constant multiple of A_N. Thus no coefficient vanishes for a positive parameter, and the support counts hold for every $N \geq 1$. Let C_0 be the largest absolute constant coefficient. Its exact integer value is recorded in the certificate and has 1520 binary digits. The largest absolute coefficient multiplying A_N is

$$L_0 = 2 \cdot 80989^2 = 13118436242.$$

The complete integer comparison verifies

$$80989^{94} + 1 > C_0, \qquad 80989^{47} > L_0, \qquad 80989^{90} + 1 < C_0.$$

Consequently, for both families,

$$H(P_N) = 80989^{4N+2} + 1 \qquad (N \geq 23).$$

Indeed this coefficient exceeds both C_0 and $L_0 A_N$ at $N = 23$, and continues to do so as N increases. Since $C_0 > L_0^2$, the exact height for any positive parameter is $\max(C_0, A_N^2 + 1)$. This is an algebraic height threshold; the sufficient parameter in the 342-variable defining theorem remains 27464, and the corresponding threshold for the 297-variable family remains ineffective.

5 Projection proofs

The point, scalar, formal-logarithm and height arguments now apply with exactly their original local orders. We give the numerical deductions explicitly. Write the nonzero integer root labels as n, a, t, b, and put $B = \max(|n|, |a|, |t|, |b|)$ and $c = |a - 1|$. The retained scalar construction and its exceptional-point exclusions imply

$$c \geq 80989^{L_N}, \qquad L_N = \left\lfloor \frac{187(2N-1)}{8} \right\rfloor + 1, \qquad h(\beta) \leq \mathcal{H} c^{19/24}.$$

Its equations are unchanged. The common-place conclusion of Proposition 2.3 supplies the orders required by the formal-logarithm argument. Thus the fortieth powers of every contact radical divide

$$\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1.$$

If $n - \beta \neq 0$, the tenth powers divide its nonzero reduced numerator J, which satisfies $|J| \leq 2B\exp(h(\beta))$. These are the precise arithmetic comparison conclusions used below; no new interpretation of an unguarded point or ratio is introduced.

For three scales, the proved effective height theorem gives

$$B^2 \leq 2h(S_0) + 2h(S_0 + S_1) + 82.$$

If the kth powers of all three radicals divide a nonzero integer J, the contact-height theorem bounds the heights of $\lambda, S_0^2\lambda$ and $(S_0 + S_1)^2\lambda$ by $T = (\mathcal{H} + \log 3)|J|^{19/k}$. Since $\lambda \neq 0$, both $h(S_0)$ and $h(S_0 + S_1)$ are at most T. As $\mathcal{H} > 92$, we obtain

$$B^2 \leq 4T + 82 \leq 5\mathcal{H}|J|^{19/k}. \tag{7}$$

For $N \geq 27464$, the independently verified fixed-prime estimates give

$$|a| > (40\mathcal{H})^{10}, \qquad \log(48\mathcal{H}) < |a|/40, \qquad h(\beta) < |a|/40.$$

If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$. Equation (7) with $k = 40$ yields $B^2 < 20\mathcal{H}B^{19/10}$, contradicting $B \geq |a|$. Therefore $\Delta = 0$. The elementary integral Lucas descent gives $\log B > |a| - 3$. If $n - \beta \neq 0$, equation (7) with $k = 10$ implies

$$\log B < 10\log(20\mathcal{H}) + 19h(\beta) < 29|a|/40,$$

a contradiction. Thus $n = \beta \in \mathbb{Z}$.

For two scales, the aggregate trace-height theorem supplies constants $M_1 > 0, M_2 \geq 1$ with

$$B^2 \leq M_1 h(S_0) + M_2.$$

The additive constant is not effective in that proof. The same height calculation for $\lambda, S_0^2\lambda$ gives

$$B^2 \leq C|J|^{19/k} + M_2, \qquad C = M_1(\mathcal{H} + \log 3). \tag{8}$$

If $\Delta \neq 0$, this implies $B^2 \leq C14^{19/40}B^{19/10} + M_2$, impossible for sufficiently large B. The scalar lower bound ensures $B \geq c - 1 \to \infty$ as $N \to \infty$. Hence $\Delta = 0$ for all sufficiently large N. For $B^2 \geq 2M_2$, if $n - \beta \neq 0$, (8) gives

$$\log B \leq 10\log(2C) + 19\log 2 + 19h(\beta) \leq 10\log(2C) + 19\log 2 + 19\mathcal{H}c^{19/24} = o(c),$$

contrary to the Lucas lower bound $\log B > c - 4$. Therefore $n = \beta$ also in the two-scale system for every sufficiently large N.

In either case, writing $X = u/v$ in lowest terms, $v > 0$, the integrality of $51X^2 + 2504$ implies $v^2 \mid 51$. Since 51 is squarefree, $v = 1$. This proves soundness. Proposition 3.2 and the retained root and scalar preparation prove completeness for every integer input. Over $\mathbb{Q}$ a sum of squares vanishes exactly when every quadratic row vanishes. These observations prove the projection assertions in Theorem 1.1.

6 Coefficient verification and reproduction

The producer `build_quartic_shared_weights.py` reads the pinned complete source tables, performs the displayed row changes, and expands every square by exact integer arithmetic. The independent checker `check_quartic_shared_weights.py` does not import the producer. It reconstructs every changed row from (6), compares every retained row, and expands the entire squared sum using unordered pairs of monomials. It also checks all coordinate incidences, the input coordinate, degree four and content one. No coefficient is sampled.

Before fixed-parameter powers are collected, the quartic tables contain 4269 and 4939 terms. Only the coefficient A^2+1 of r^2 mixes parameter degrees; every other coefficient is a nonzero constant or a nonzero multiple of A. Thus their supports are constant for $A>0$. The coefficient classification and exact integer comparisons in the preceding normalization proof give the height formula for $N \geq 23$. Direct evaluation at $A = 80989^{54929}$ gives 1791283 binary digits for the largest coefficient. The checks also retain the elementary uniform bound $H(P_N) < 2^{1520} A_N^2$ for all $N \geq 1$.

The finite verification includes exact rational substitutions in the new weight chain and 61824 bounded valuation checks. These supplement Proposition 2.3, whose proof covers every odd valuation; they are not a substitute for that proposition or the arithmetic premises. The two reproduction programs use only the Python standard library.

References

[1] *Separation and denominator conditions in quadratic prime patterns*, 9 November 2026. Source: `prime_selection_20261109_v12.tex`.

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[3] *A forced prime condition and an explicit quartic in 392 variables*, 8 November 2026. Arithmetic input contract, fixed-prime certificate, scalar exclusions, and Lucas descent. Source: `quartic392_20261108_v8.tex`.

[4] *Aggregate trace heights of elliptic multiples*, 8 November 2026. Source: `aggregate_trace_height_20261108_v9.tex`.

[5] *Effective heights of elliptic abscissa sums*, 8 November 2026. Source: `effective_elliptic_sums_20261108_v10.tex`.

[6] *An explicit quartic in 351 variables*, 9 November 2026. Complete quadratic source table with SHA-256 `452d063b6d7da598471aa1ec0302b9e37611a44168610592931ec56aac241966`.

[7] *A family of quartics in 301 variables*, 9 November 2026. Complete quadratic source table with SHA-256 `9917be941135778816ce25ecb0cbc872b4ed7c9f3691d79355e5e8352667a4fe`.

[8] E. Glazer, *An explicit quartic defining the integers in the rationals*, Version 13, manuscript and coefficient data, commit `5bbd74f6a479776a332cbe036809d5dddae63720`, https://github.com/elliotglazer/diophantine-z-q.